CEN-300 Practical Problem

Spatio-Temporal Vehicle Trajectory Prediction and Conflict Detection Using Attention based BiLSTM

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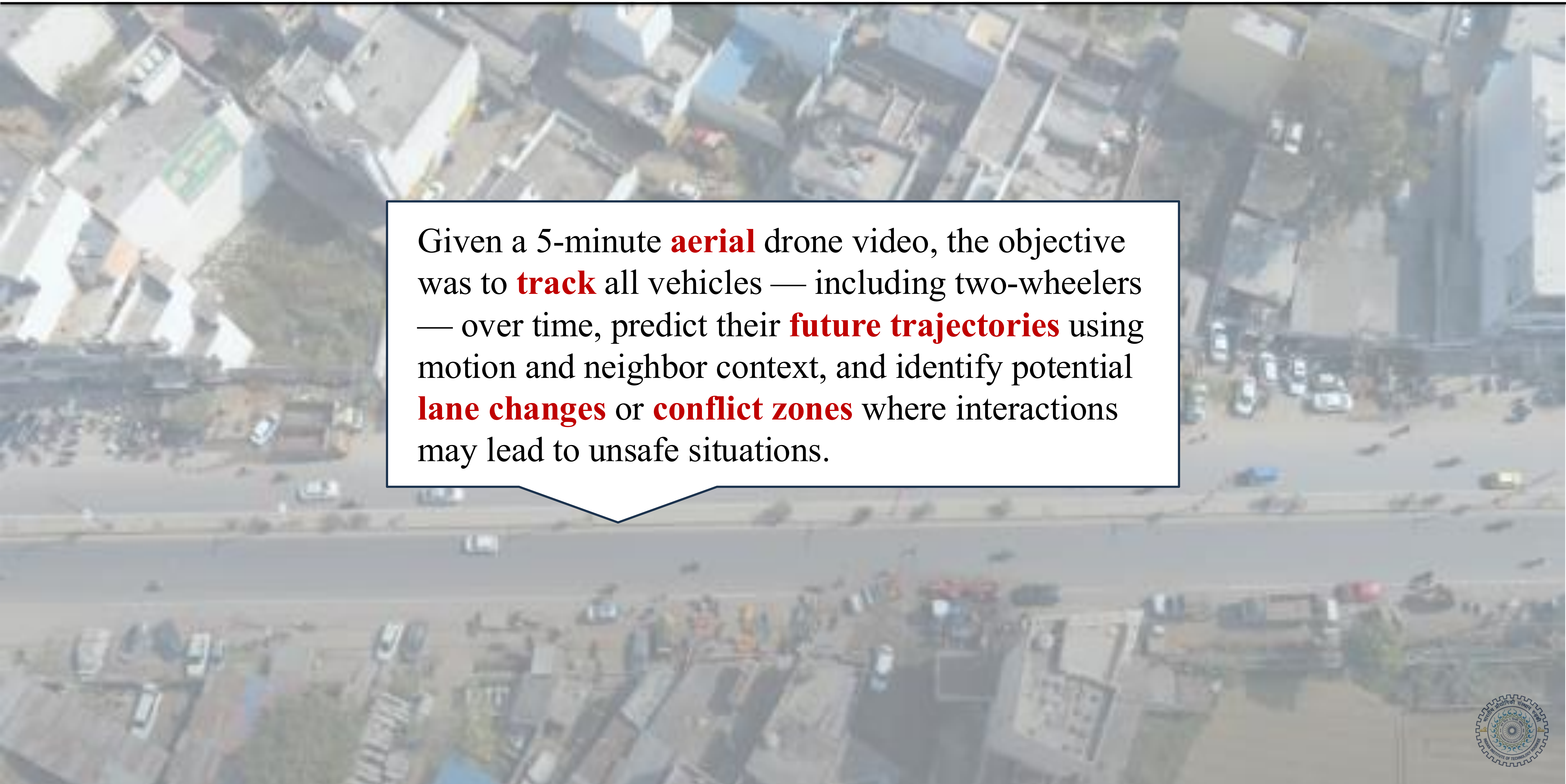
Problem Statement – Vehicle trajectory prediction and conflict zone mapping



Given a 5-minute **aerial** drone video, the objective was to **track** all vehicles — including two-wheelers — over time, predict their **future trajectories** using motion and neighbor context, and identify potential **lane changes** or **conflict zones** where interactions may lead to unsafe situations.

Project Timeline – Jan-May 2025.

Week 1-3: Read Research Papers, had discussions and decided solution.

Week 4-6: Implemented Object Detection Tracking and trained Yolov8s on custom annotated dataset.

Week 7-8: Trained and implemented LSTM Kalman filter algo for vehicle trajectory prediction.

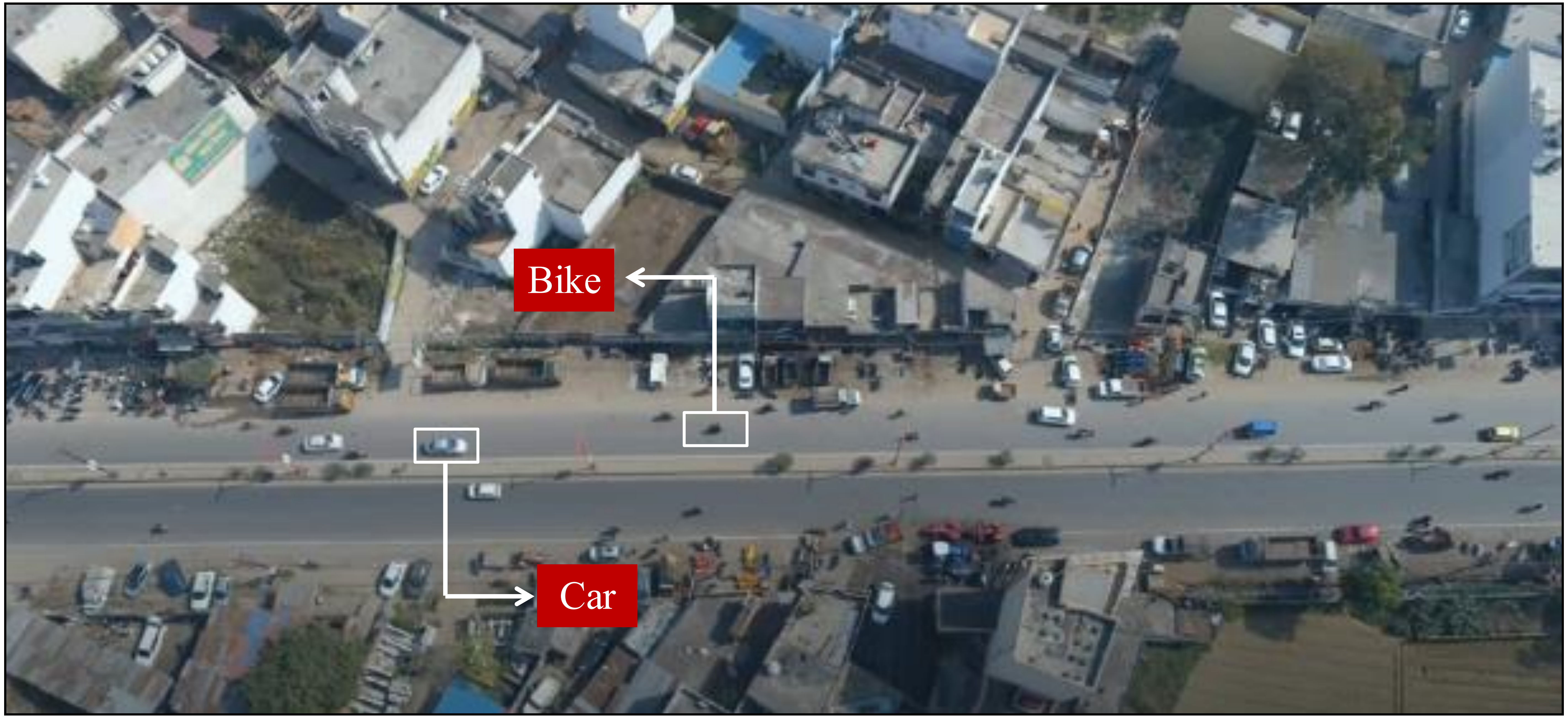
Week 9-11: Coded and trained SDT-ATT model from scratch on 30s video sample for multivariate vehicle trajectory.

Calculated PET and determined Conflict Zones from the predicted trajectory values.

Week 12-13: Report, Platform and presentation finalised.

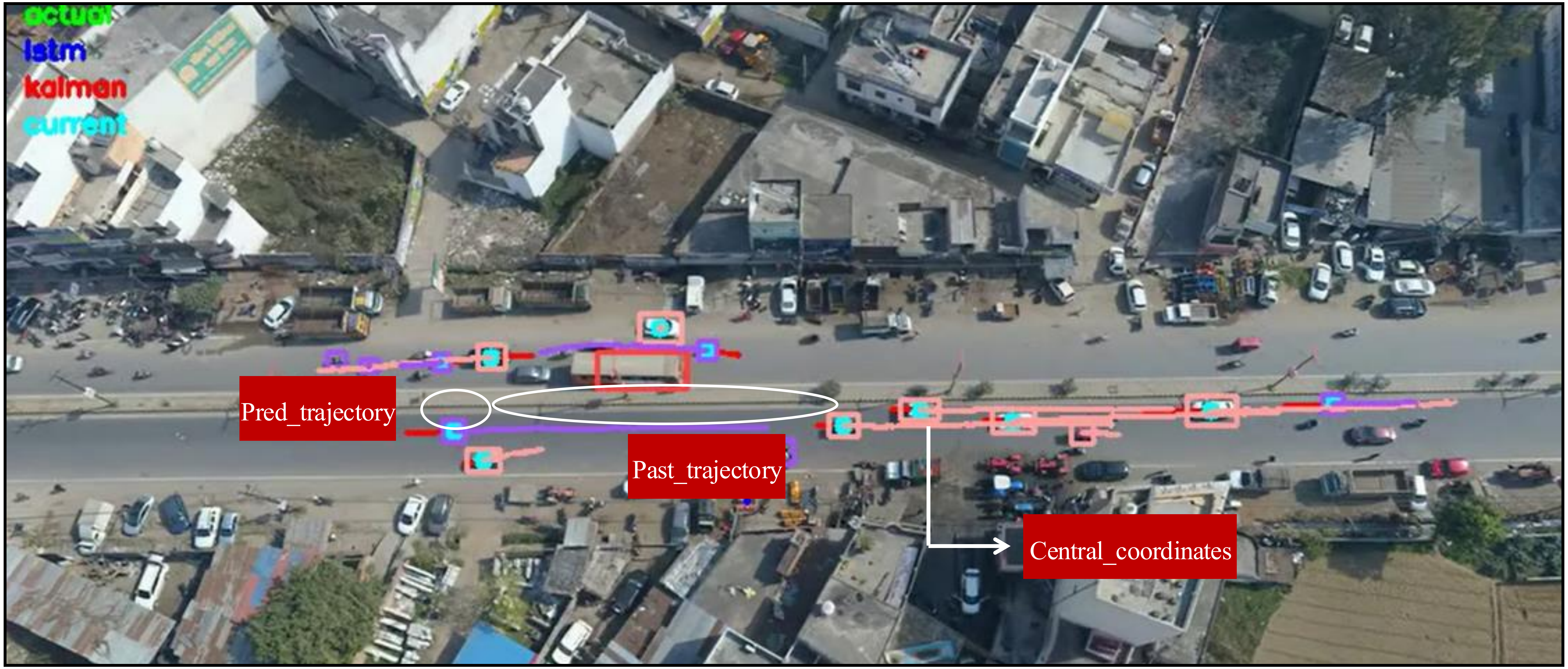
Solution Workflow

1 Vehicle Detection using 5min aerial drone footage



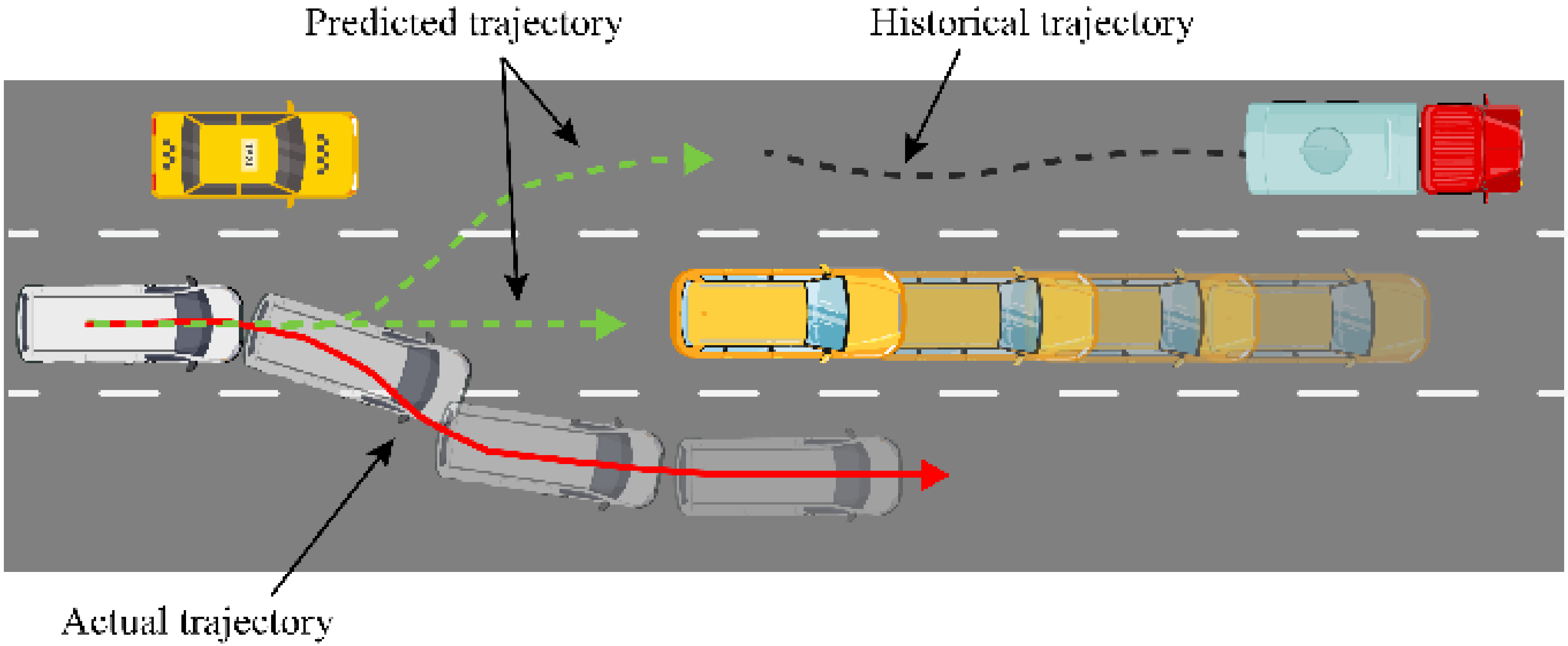
Solution Workflow

2 Trajectory Extraction & Mapping



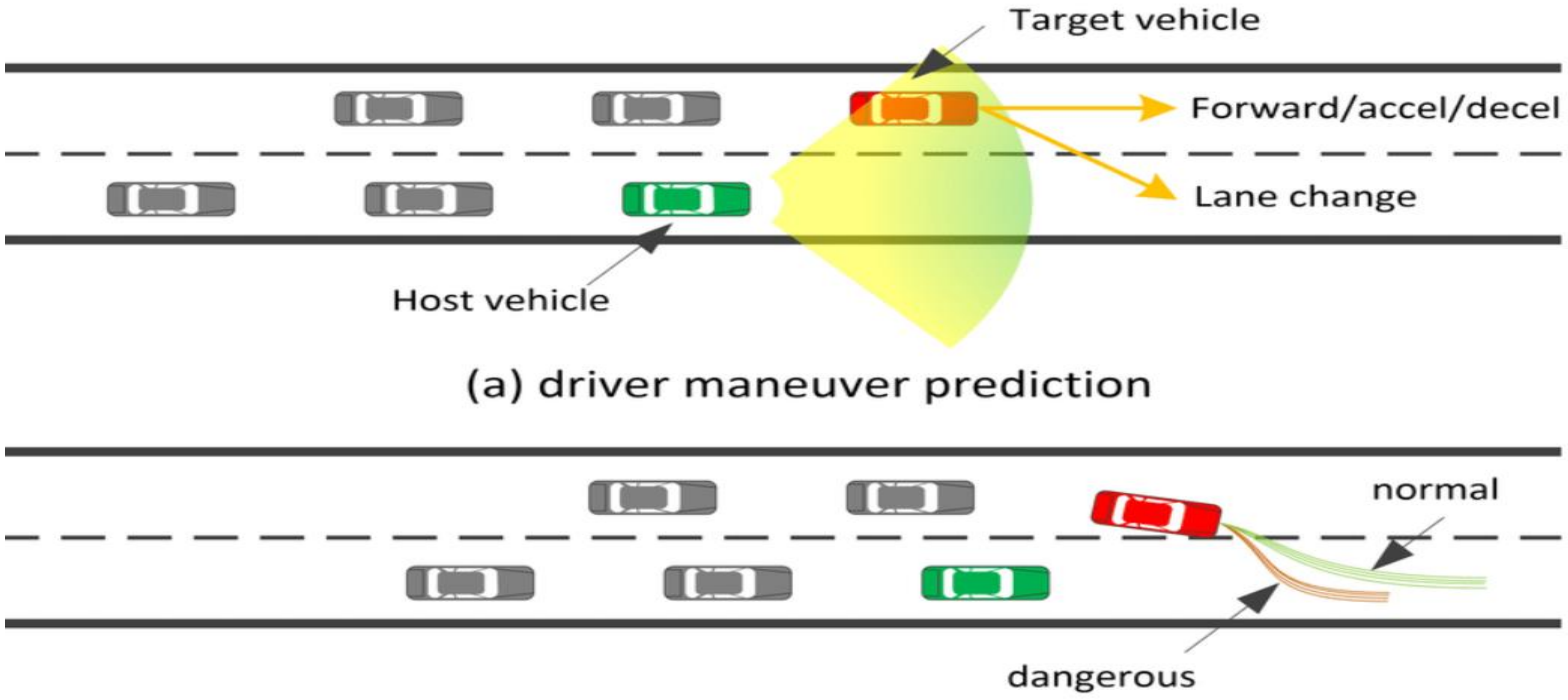
Solution Workflow

3 **Future trajectory prediction using DL + Physics-based models** (from [Paper])



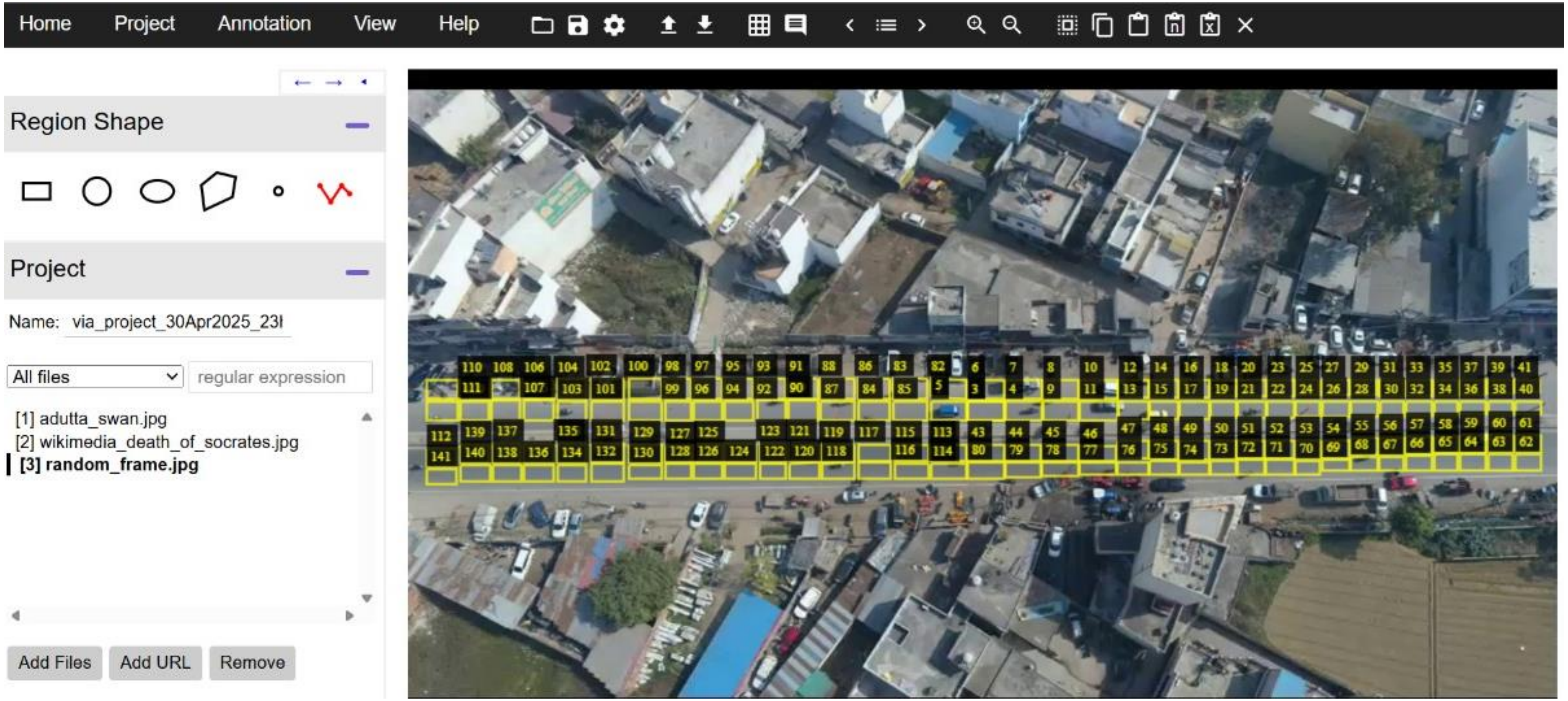
Solution Workflow

4 Lane Change Prediction based on custom algorithms and logic



Solution Workflow

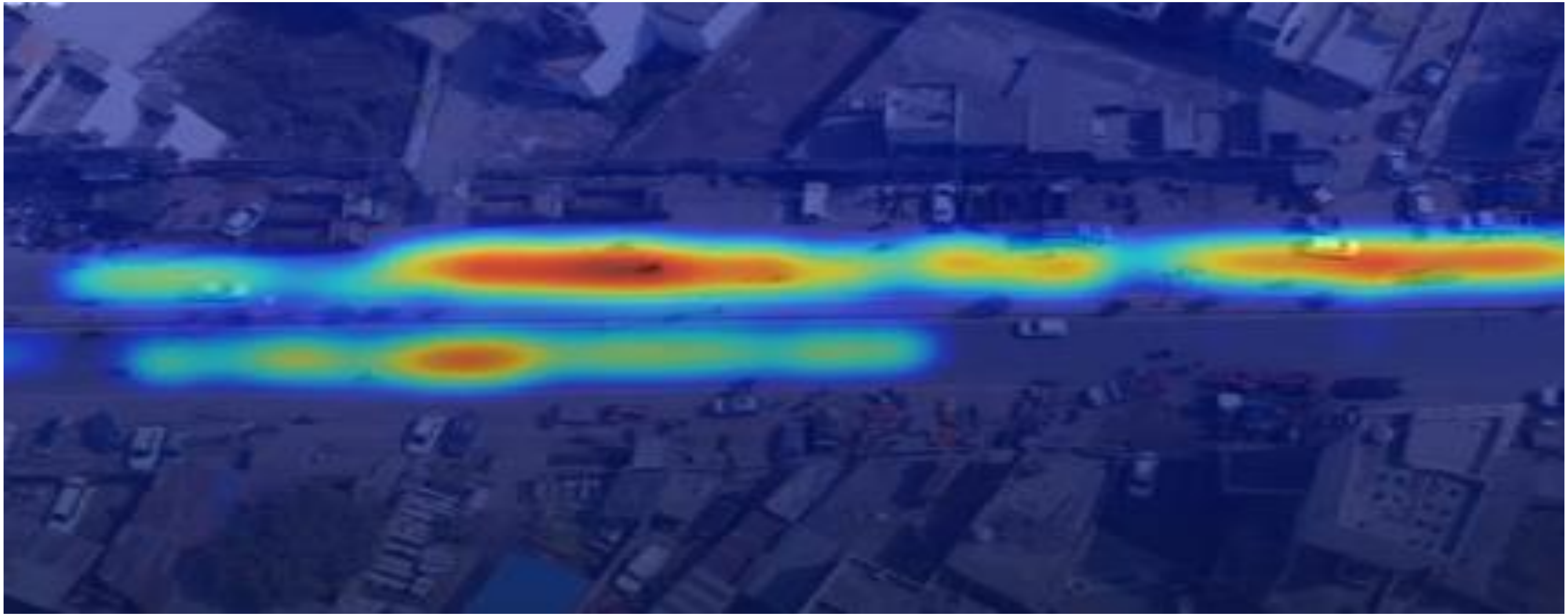
5 Estimation of potential collision zones by calculating Post-Encroachment Time (PET).



Using VIA Annotator

Solution Workflow

6 **Categorization of regions into different risk zones based on collision prediction outcomes.**



Libraries, Algorithms and Concepts involved in the project

PyTorch

pandas

ultralytics YOLO

OpenCV

NumPy

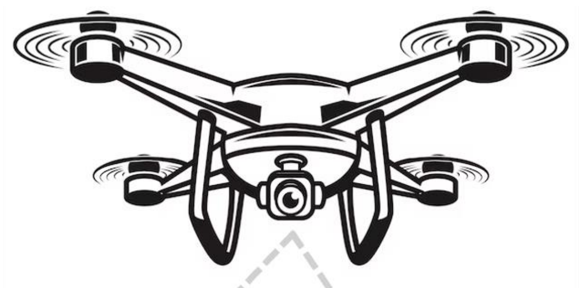
roboflow

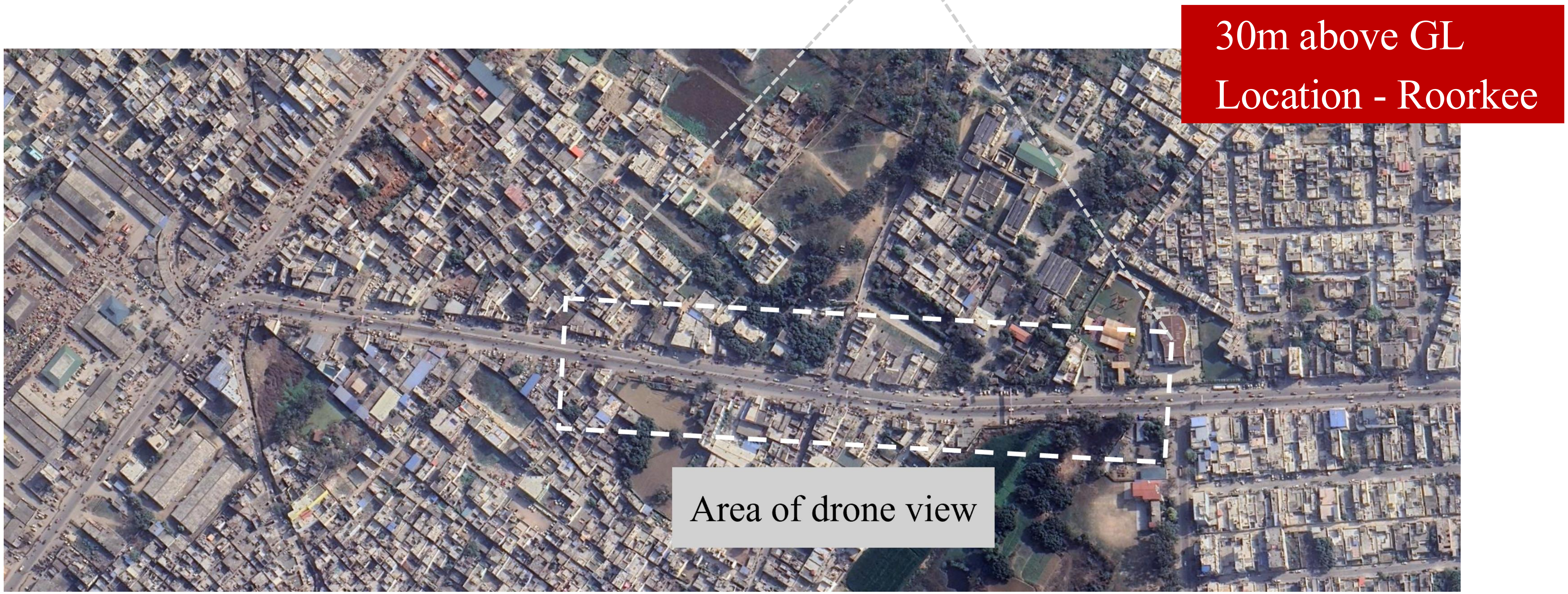
seaborn

Data processing and annotations

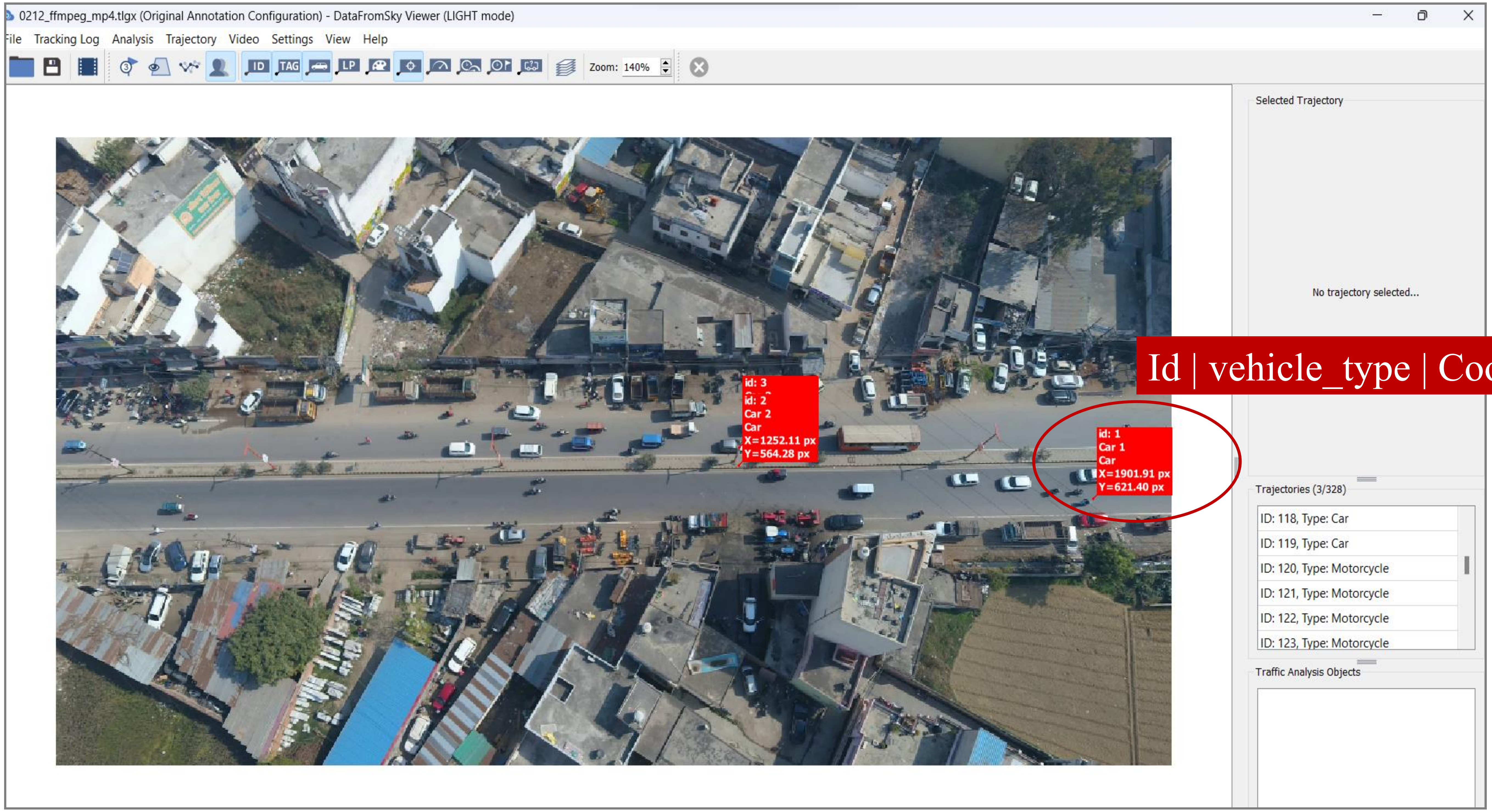
Step 1: Vehicle detection and Tracking from Aerial drone video

DATA COLLECTION





Dataset processing using Data-from-sky, and problems with it



Id | vehicle_type | Coordinates

Exported data having unstructured data

	A
1	Track ID; Type; Entry Gate; Entry Time [s]; Exit Gate; Exit Time [s]; Traveled Dist. [m]; Avg. Speed [km/h]; Trajectory(x [m]; y [m]; Speed [km/h]; Tan. Acc. [ms-2]; Lat. Acc. [ms-2]; Time [s];)
2	1; Car; - ; - ; - ; - ; 75.11; 14.627646; 779538.37; 3308327.73; 6.4299; 0.0770; 0.0077; 0.000000; 779538.36; 3308327.76; 6.4337; 0.0514; 0.0085; 0.016683; 779538.35; 3308327.79; 6.4374; 0.0718; -0.0
3	0.3000; 9.292617; 779524.82; 3308355.94; 18.9997; 0.2732; 0.3263; 9.309300; 779524.78; 3308356.02; 19.0166; 0.2909; 0.3374; 9.325983; 779524.73; 3308356.10; 19.0326; 0.2406; 0.3367; 9.342667
4	3308394.56; 3.4953; -3.5492; -0.3307; 18.385033; 779504.68; 3308394.56; 3.3882; -3.5543; -0.3416; 18.401717; 779504.68; 3308394.56; 3.2799; -3.5903; -0.3058; 18.418400; 779504.68; 3308394.56;
5	2; Motorcycle; - ; - ; - ; - ; 68.21; 7.578674; 779547.17; 3308317.48; 7.1780; -0.0655; -0.1168; 0.000000; 779547.14; 3308317.46; 7.1733; -0.0920; -0.0631; 0.016683; 779547.11; 3308317.44; 7.1691; -0
6	812; -0.0620; 9.325983; 779534.09; 3308306.79; 1.1511; -0.1598; -0.0123; 9.342667; 779534.09; 3308306.79; 1.1413; -0.1659; -0.0297; 9.359350; 779534.09; 3308306.79; 1.1319; -0.1479; -0.0280; 9.3
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10	4; Car; - ; - ; - ; - ; 61.66; 3.655082; 779518.18; 3308291.94; 15.5947; 0.3393; -0.0721; 0.000000; 779518.22; 3308291.99; 15.6151; 0.3418; -0.0247; 0.016683; 779518.27; 3308292.05; 15.6337; 0.2766;
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ID1

ID3

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.
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uhyuh

Tried formatting into desired format, but replicated values observed for same tracking IDs

Data inconsistencies in Data From Sky for every track ID

Change in data format for Car 1

553	1	Car	-	-	-				.13	3308355.4	18.8805	0.2914	0.3165	9.192517
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557	1	Car	-	-	-	-	75.11	14.627646	779524.95	3308355.7	18.9508	0.2654	0.3475	9.259250
558	1	Car	-	-	-	-	75.11	14.627646	779524.91	3308355.7	18.9670	0.2749	0.3163	9.275933
559	0.3000	9.292617	779524.82	3308355.9	18.9997	0.2732	0.3263	9.309300	779524.78	3308356.0	19.0166	0.2909	0.3374	9.325983
560	0.3000	9.292617	779524.82	3308355.9	18.9997	0.2732	0.3263	9.309300	779524.73	3308356.1	19.0326	0.2406	0.3367	9.342667
561	0.3000	9.292617	779524.82	3308355.9	18.9997	0.2732	0.3263	9.309300	779524.69	3308356.1	19.0492	0.3110	0.3036	9.359350
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568	0.3000	9.292617	779524.82	3308355.9	18.9997	0.2732	0.3263	9.309300	779524.38	3308356.7	19.1643	0.2824	0.3467	9.476133
569	0.3000	9.292617	779524.82	3308355.9	18.9997	0.2732	0.3263	9.309300	779524.34	3308356.7	19.1804	0.2544	0.3212	9.492817
570	0.3000	9.292617	779524.82	3308355.9	18.9997	0.2732	0.3263	9.309300	779524.30	3308356.8	19.1959	0.2622	0.3393	9.509500
571	0.3000	9.292617	779524.82	3308355.9	18.9997	0.2732	0.3263	9.309300	779524.25	3308356.9	19.2117	0.2647	0.3669	9.526183
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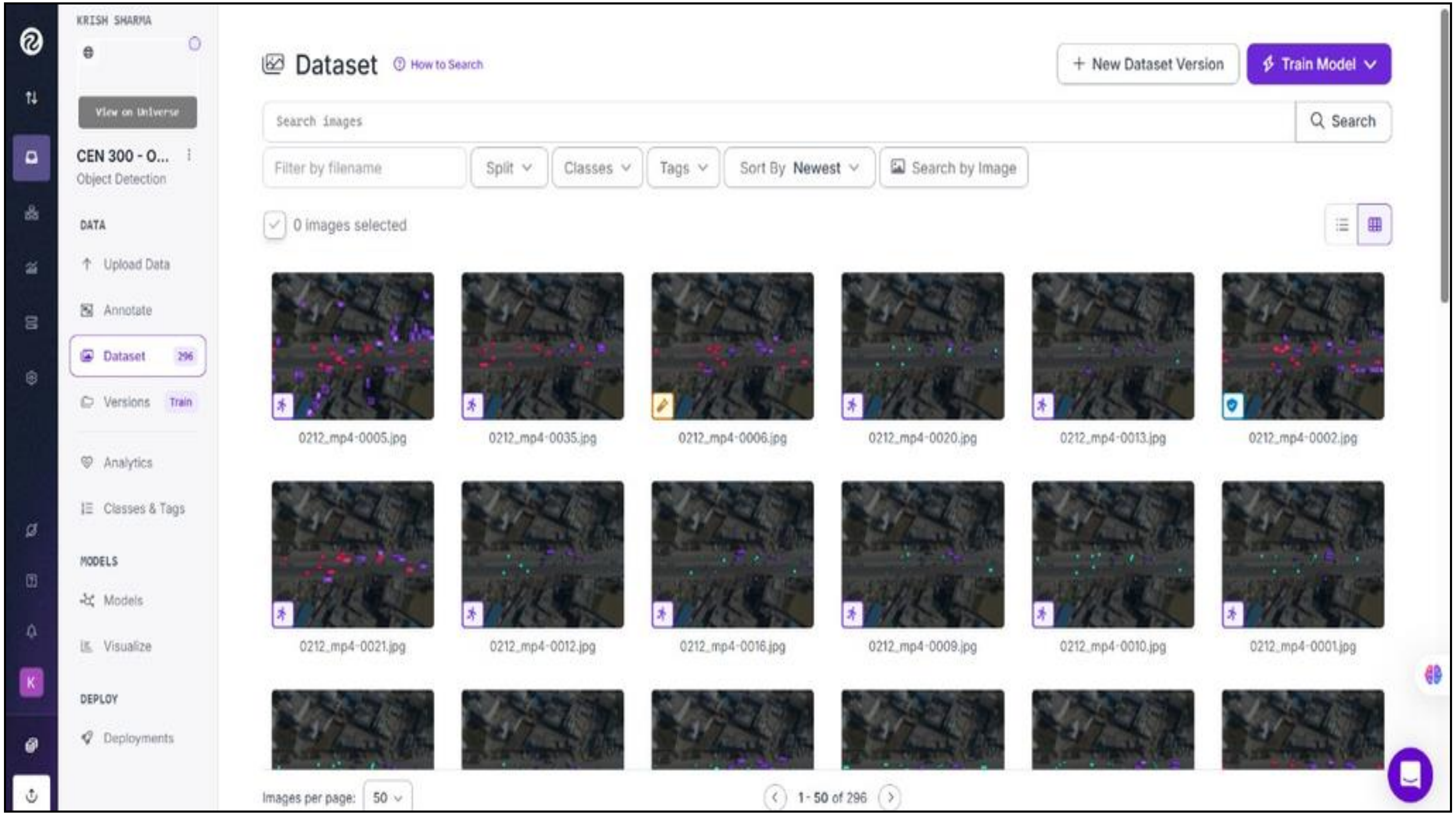
573	0.3000	9.292617	779524.82	3308355.9	18.9997	0.2732	0.3263	9.309300	779524.17	3308357.1	19.2448	0.2602	0.3777	9.559550
574	0.3000	9.292617	779524.82	3308355.9	18.9997	0.2732	0.3263	9.309300	779524.12	3308357.1	19.2602	0.2526	0.3099	9.576233

Annotation of vehicles using Roboflow to enhance detection performance, improve model accuracy, and include diverse vehicle types such as two-wheelers.

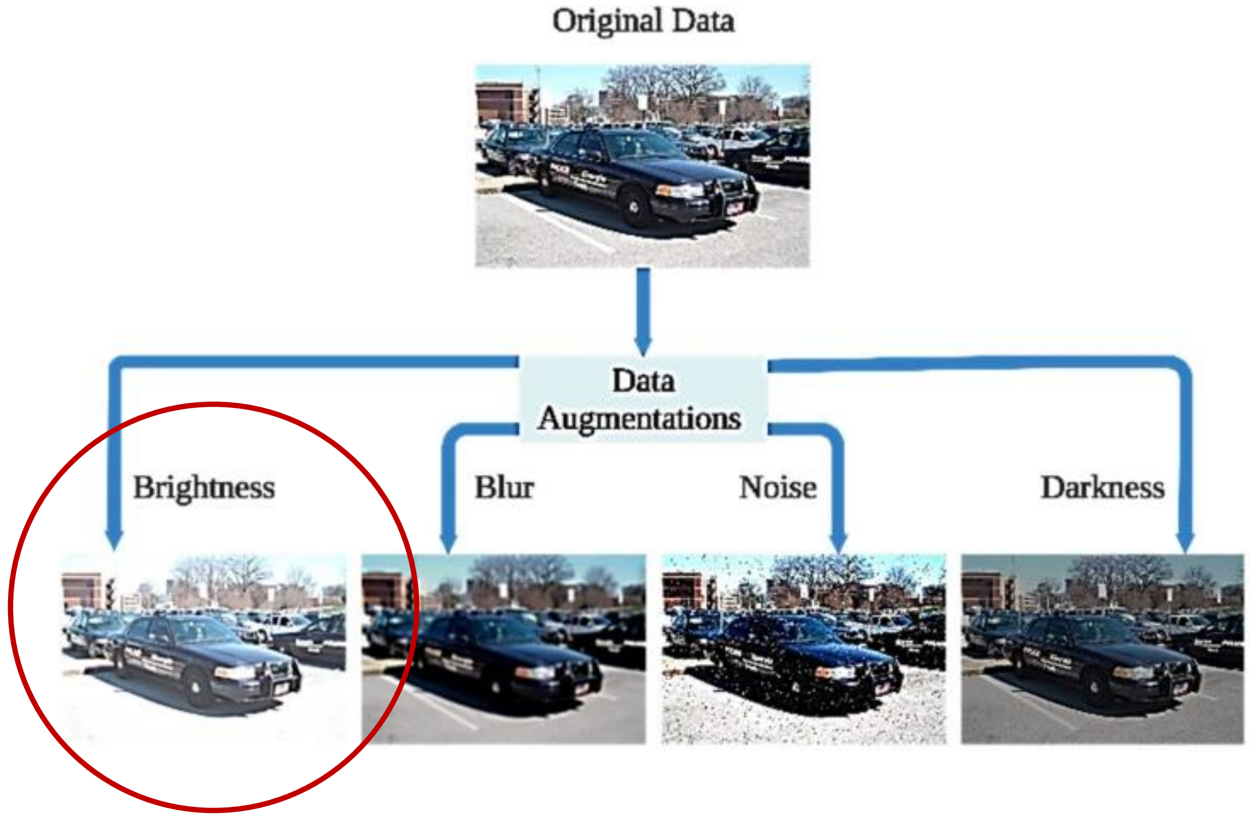


300 + images

Annotation

(car , two-wheelers , Buses)

Data Augmentation - For Upscaling test and training dataset



Brightness change + Cropping

Enhances the performance and robustness by artificially increasing the size and diversity of training data.

Rotation & blurring

Noise addition

Trajectory prediction

Vehicle detection and Tracking

Bounding Boxes
for each vehicle

* Thresholds : Confidence Score = 0.3 ; IoU = 0.7

YOLOv8

Detection

ByteTrack

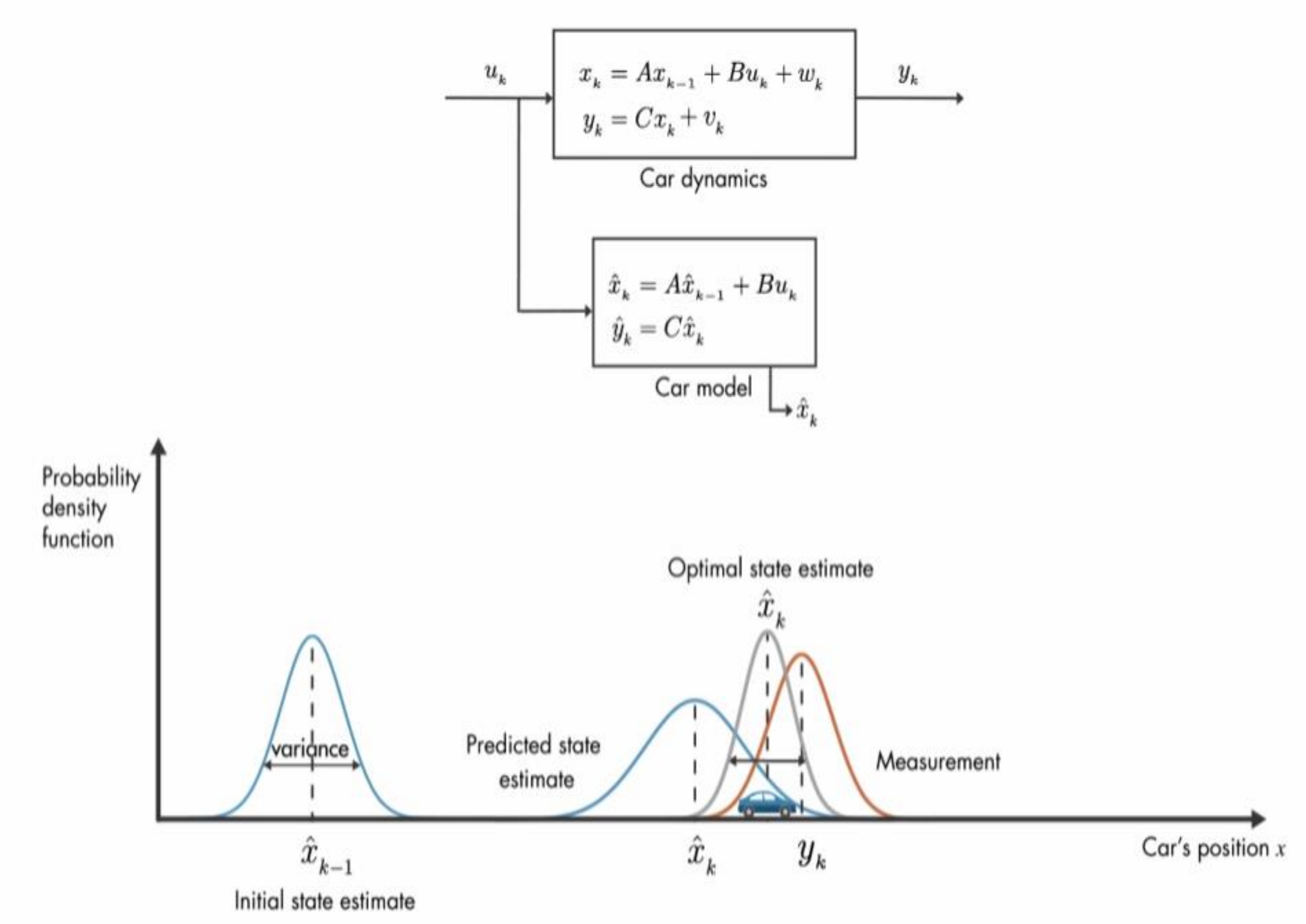
Tracking

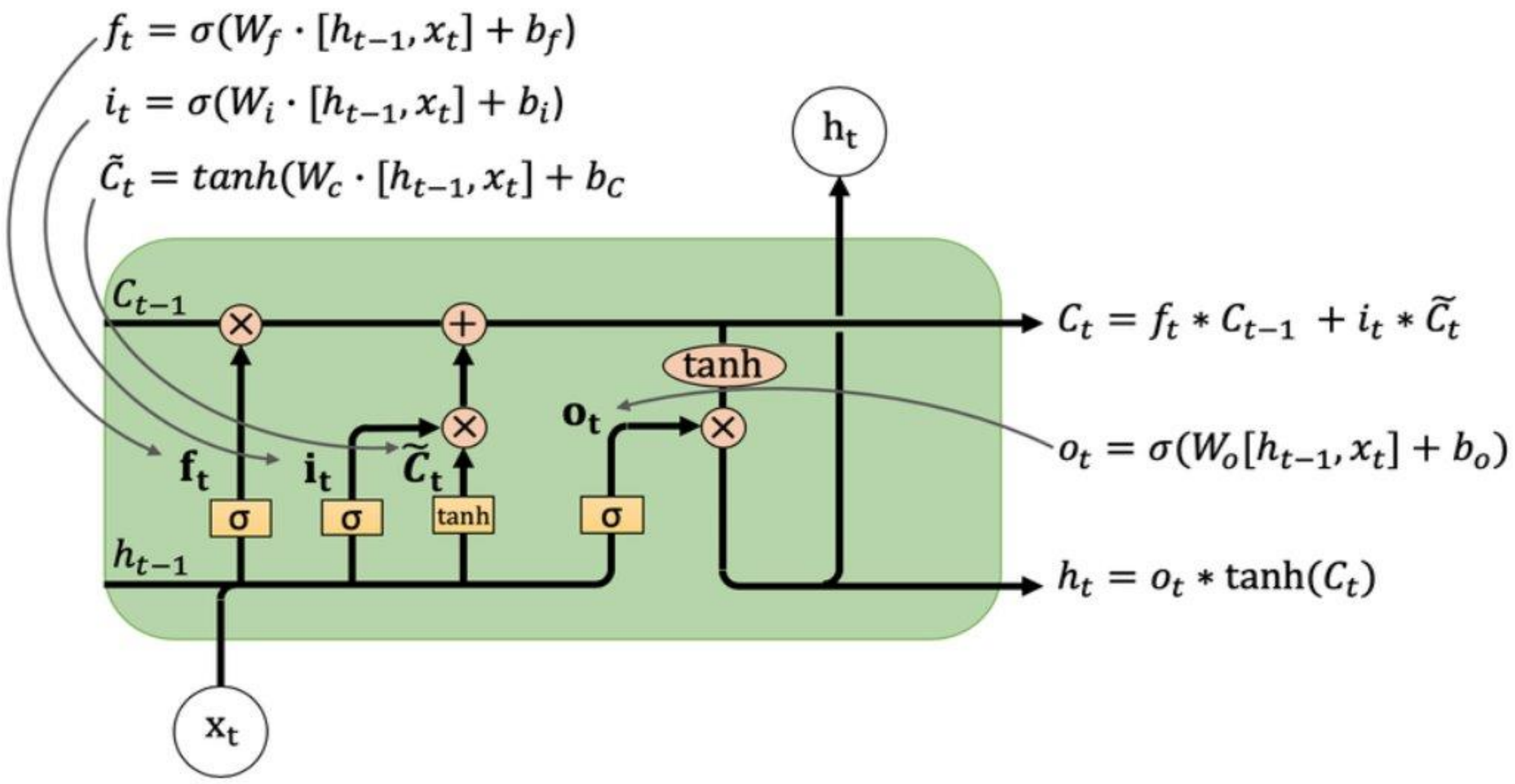
Supervision(Roboflow)

Processing detections

Model Architectures & Techniques

LSTM for motion learning, **Kalman Filter** for smoothing and velocity estimation.





Trajectory Prediction

DATA STORAGE

frame, time, tracker_id, and coordinates

timestamp (s)	tracker_id	x	y
6.022	1	232.09	756.05
6.152	1	232.13	756.03
6.259	1	232.24	756.04
6.386	1	232.20	756.00
6.474	1	232.16	755.95
6.566	1	232.30	755.87
6.662	1	232.38	755.84
6.757	1	232.39	755.78
6.845	1	232.39	755.70

....

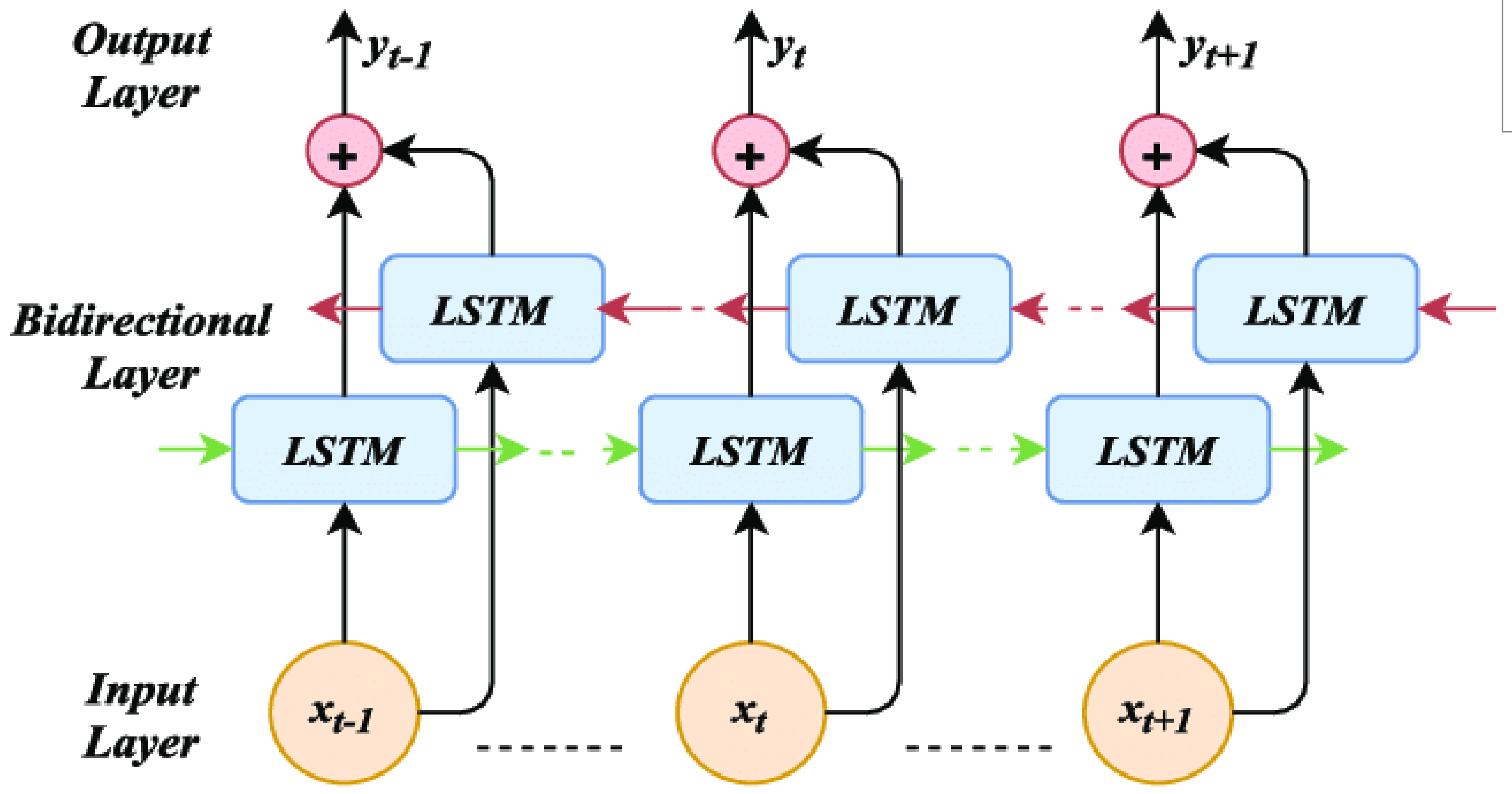
Kalman Filter (Physics-based)

Handles noisy detections & smooths trajectory

LSTM (DL-based)

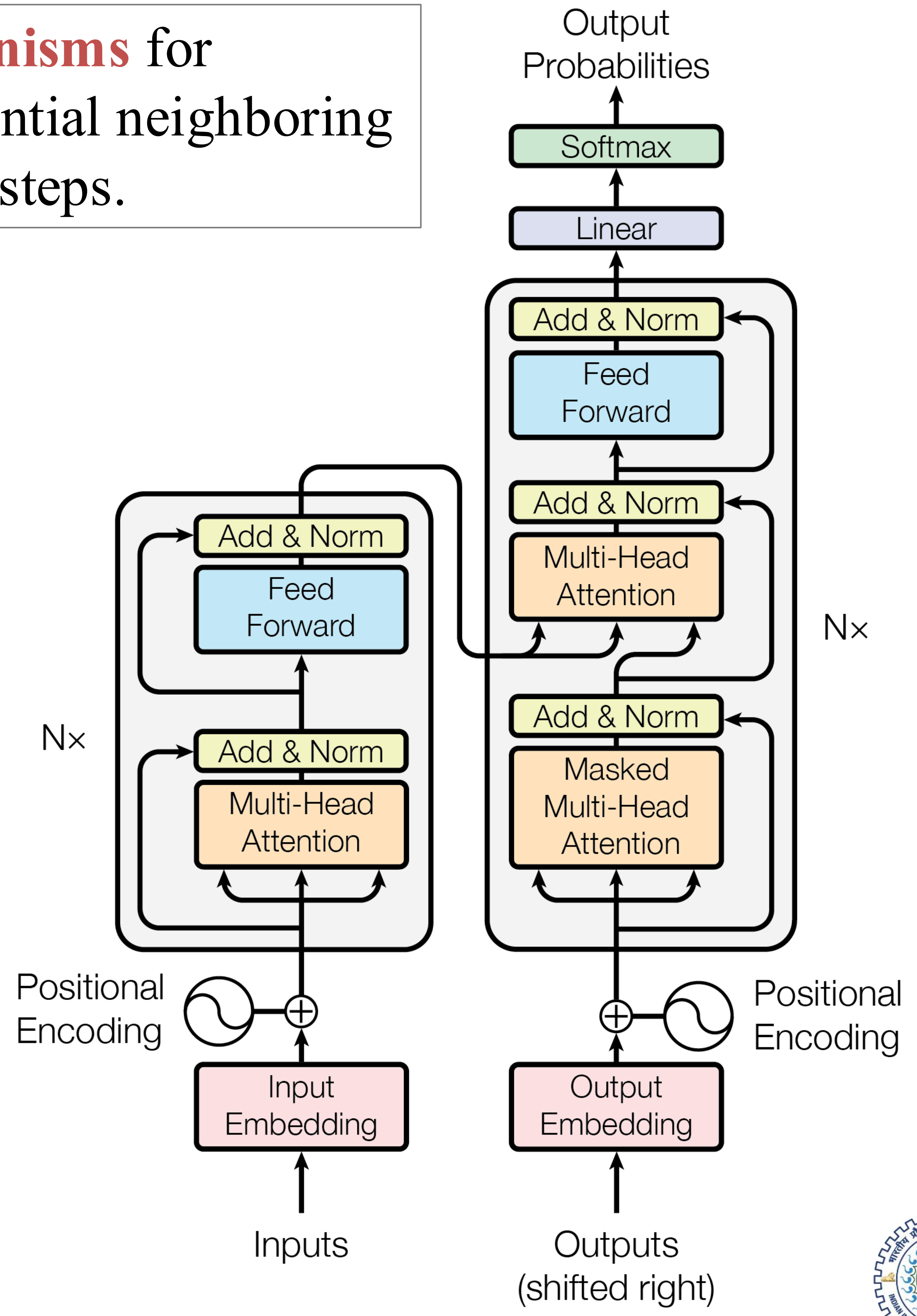
Predicts next **120 frames** for each vehicle

Model Architectures & Techniques



Bi-directional LSTMs for capturing past motion context.

Attention mechanisms for focusing on influential neighboring vehicles and time steps.



Attention-based SDT-ATT Model Implementation, for 3 neighbor and 2 lane environment

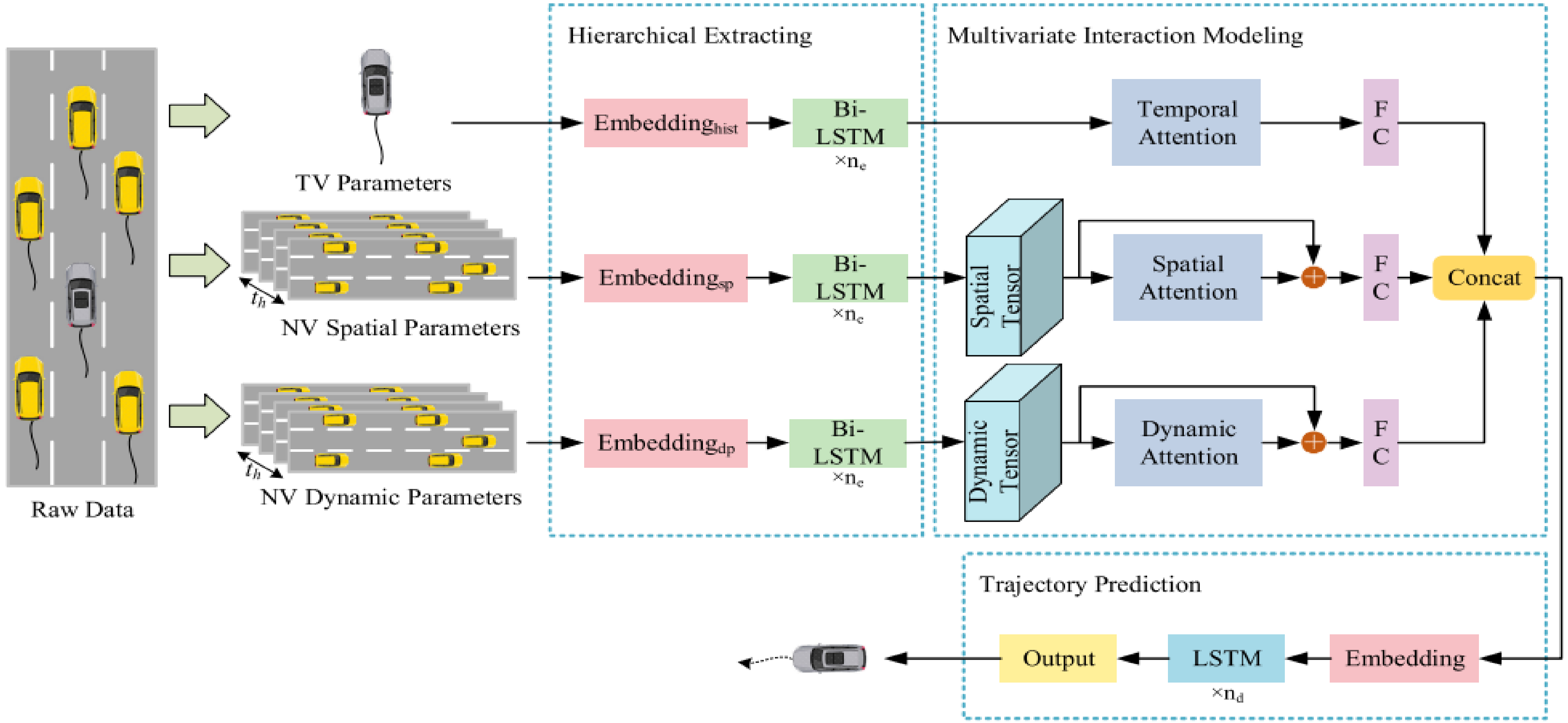
TH= 90
NF= 120

Limitations with Kalman & LSTM

- Initial model predictions had **no influence of surrounding environment.**
- This led to **inaccuracies**, as surrounding interactions significantly affect the future trajectory of a vehicle.

Solution?? → **SDT-ATT**

- Trained for 3 neighbor vehicles and 2 lane environment
- Multi-head attention & Bi-LSTM for multivariate vehicle dependencies



SDT-ATT Model Result Demonstration

Future trajectory and lane change

Vanilla SDT-ATT output were wrong for Indian Roads

Incorrect Trajectory influenced by wrong directions

Challenges and Limitations

1. Lack of standard Dataset as used in the paper. NGSIM
2. Short duration of vehicle movement
3. 2 lane multi direction drone video, contrast to the SDT-ATT paper.
4. Paper only considered single direction 3 Lane environment.
5. No Source Code available: had to code and train from scratch.

Ablation Studies

1. Training model on Cross Vehicle Trajectory Prediction
2. Cross Drone video trajectory prediction: Learning weights which can be used in different environment.
3. Real world edge device computation.

How did we send predicted trajectory to PET?

Weighted average of each frame's predicted trajectory and compared that with actual trajectory.

Trajectory Error Metrics: Error Comparison

Original paper used NGISM highway data (fixed overhead cameras, well calibrated)

Metric	Error reduction		**Paper**	Model	GPR	LSTM	Kalman	EKF
	Paper	Model	px's					
ADE	9.30%	5.20%	24.80	**25.86**	26.34	27.28	39.00	31.20
FDE	8.50%	3.70%	17.20	**18.06**	18.50	18.76	77.44	61.90

Computed on per-frame basis

Outperforms the other models

By embedding past trajectories (history), neighbour positions, and relative dynamics separately and then applying bi-LSTM, the model builds **richer context before attention**

GPR – Gaussian process regression ; LSTM – Long short-term memory ; EKF – Extended Kalman Filter

Trajectory Error Metrics: Error Comparison

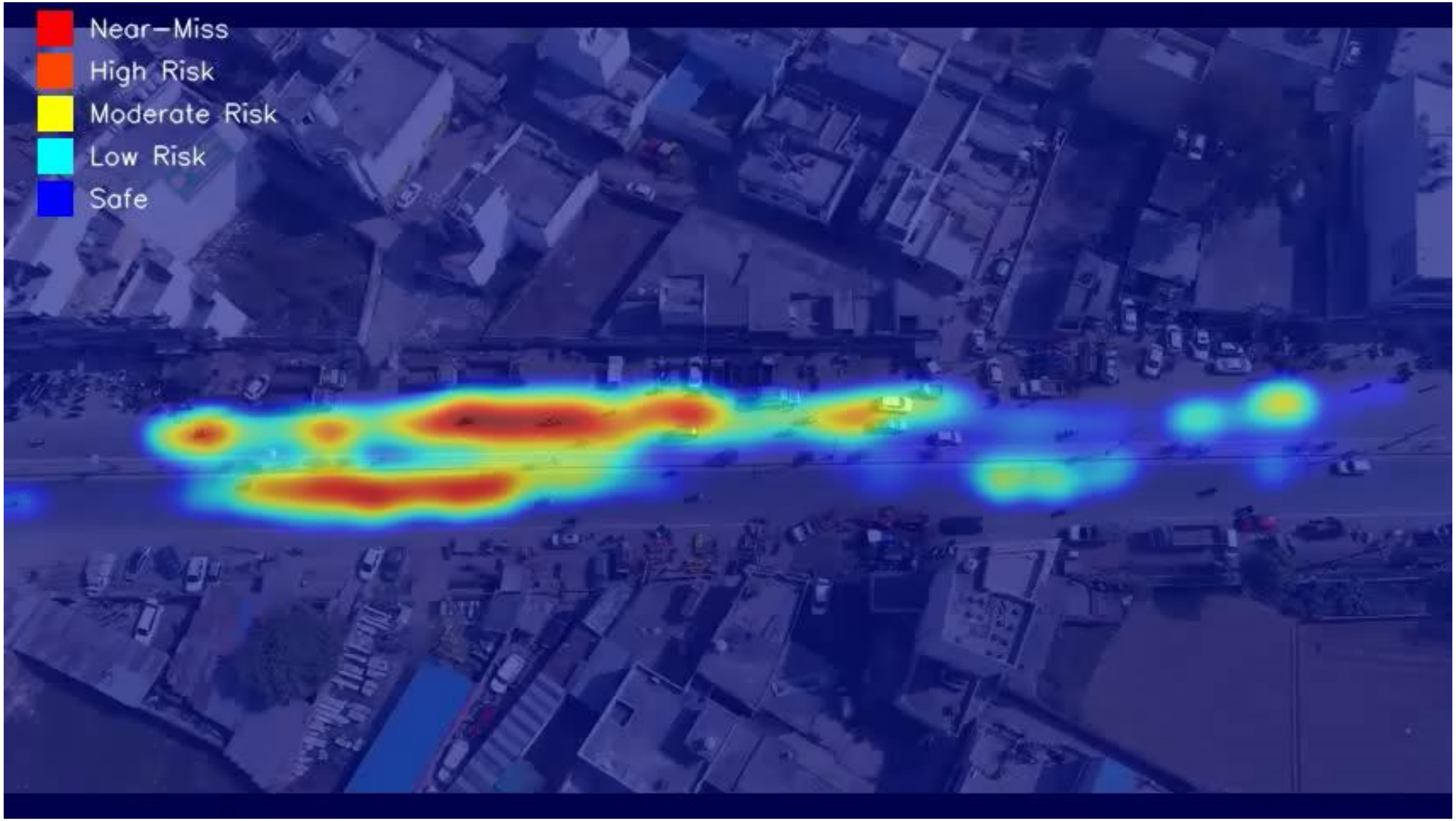
	Pros	Cons	**Result**
Gaussian Process Regression	Strong at smoothing & uncertainity quantification	Lacks learned attention to neighbour interactions	Lower performance than SDT-ATT
Extended Kalman Filter	Improves over linear KF by linearizing the motion model	Ignores social context	Lower performance than SDT-ATT
Standard LSTM	Learns sequential motion	Cannot explicitly weight spatial neighbours	Lower performance than SDT-ATT

Model Limitations & Gaps	Converting pixels from an oblique **drone angle** into a GL coordinate frame introduces scale errors
	Model performance depends on the count of **annotations**, leading to inconsistent frame intervals, missing detections

Scope of Enhancement	Incorporating **fine-grained map** priors or richer sensor fusion would mitigate viewpoint and frame-rate noise
	Multi-modal variational decoders/ graph neural variants of SDT-ATT could better capture rare maneuvers

Conflict Zone Mapping

Risk Category Heatmap based on PET values



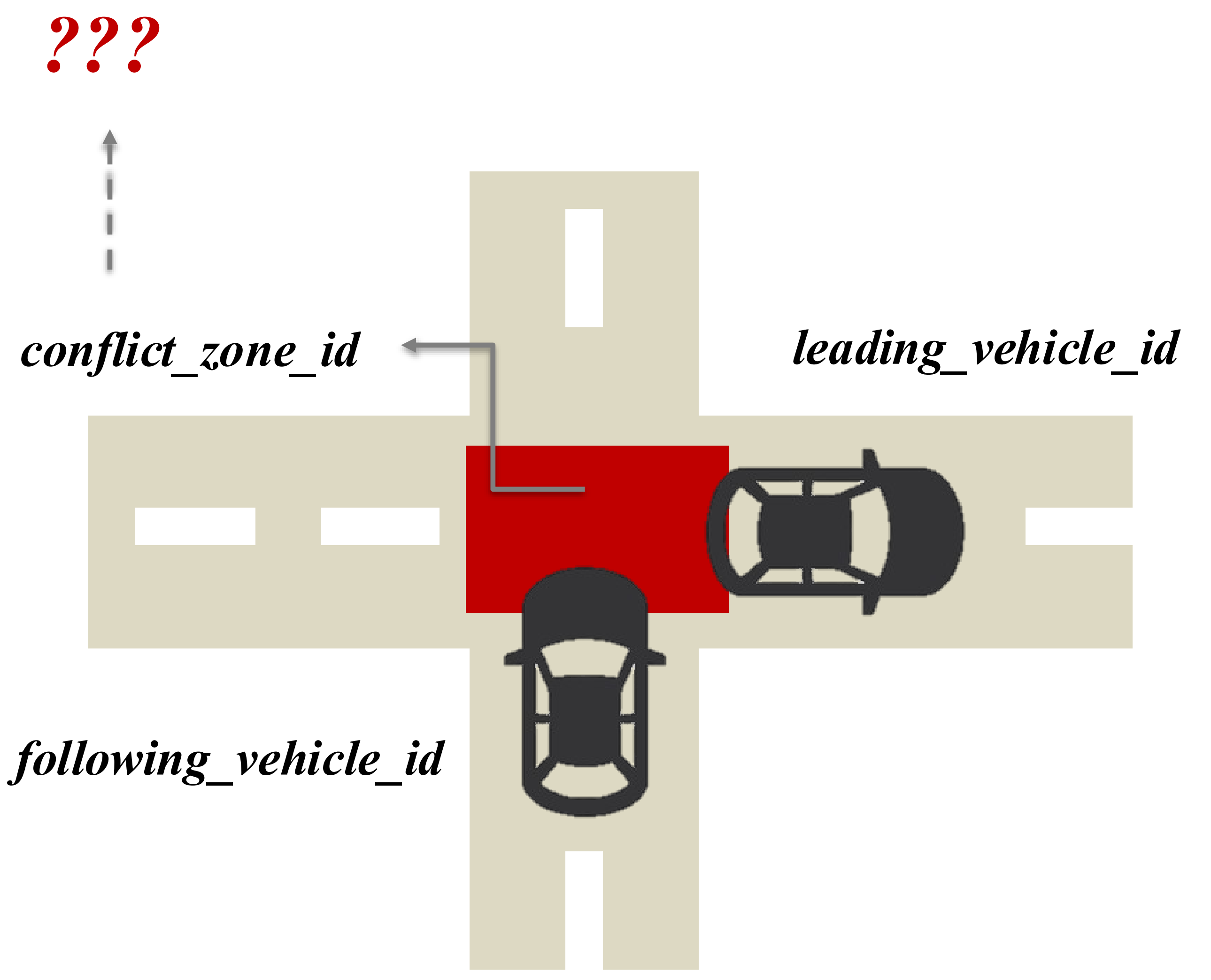
Risk Category Heatmap based on PET values

Accumulates Post Encroachment Time (PET) based on data up to a specific timestamp.

Can identify specific time when conflict zones experience more frequent high-risk events.

Computation Logic?

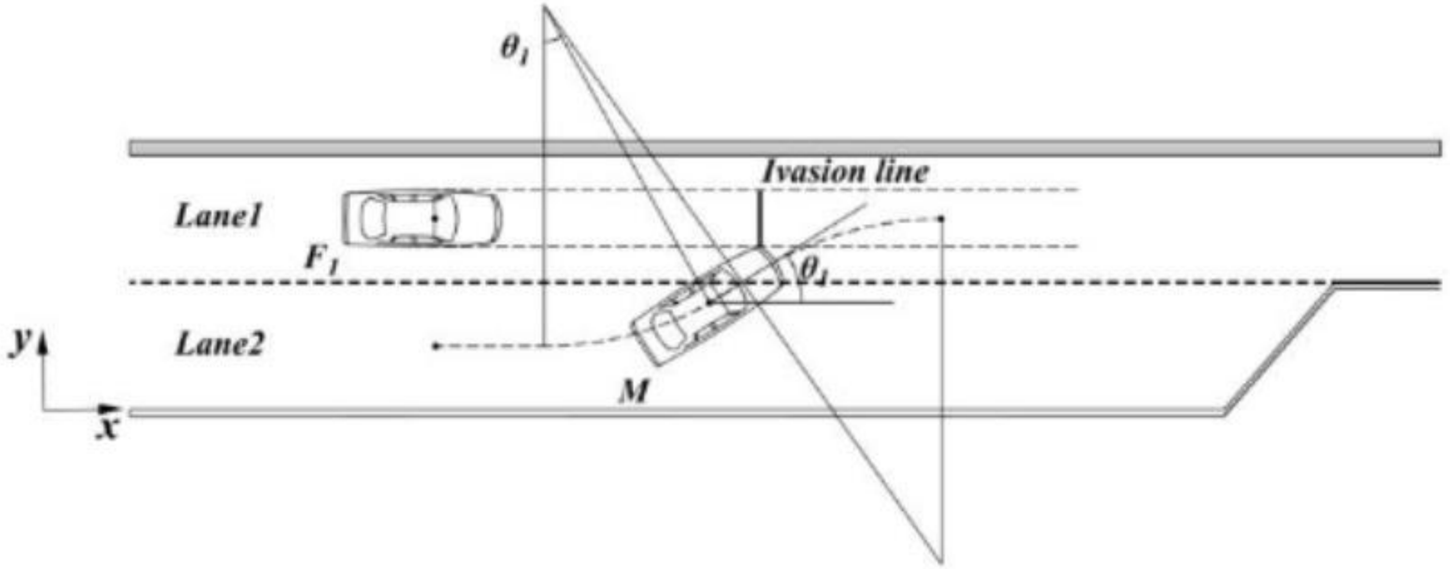
Post Encroachment Time (PET) Computation Logic



*** Inclusion/ exclusion decided using central coordinates*

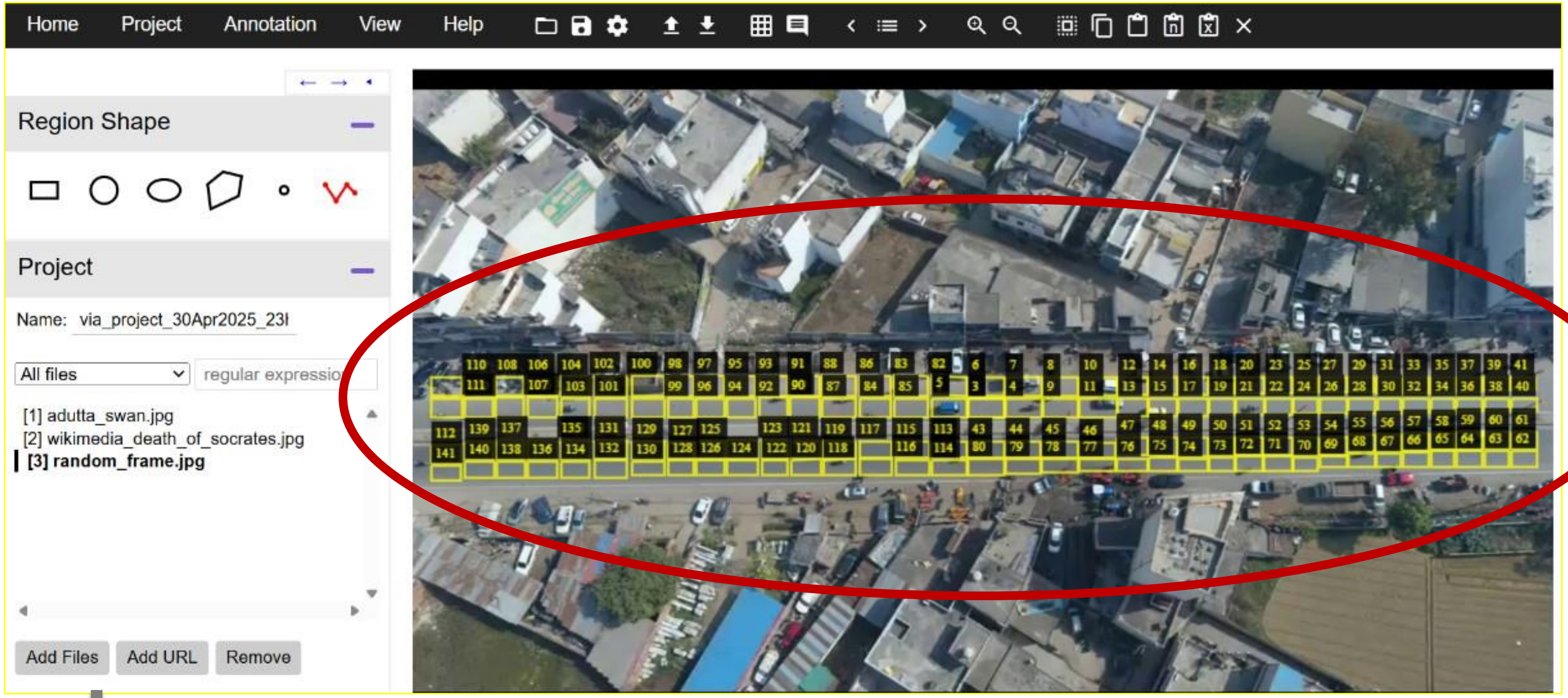
$$PET = t\_entry - t\_exit$$

$PET \leq 0$	NEAR MISS
$0 < PET \leq 1.5$	HIGH RISK
$1.5 < PET \leq 3$	MODERATE RISK
$3 < PET \leq 5$	LOW RISK
$PET > 5$	SAFE



Conflict Zone Mapping

VCG Annotator Tool



141 conflict zones mapped, divided into a grid of **2 vertical boxes per lane**

Zone_df

[29]:

	zone_id	region_shape_attributes
0	1	{"name":"polyline","all_points_x":[388,414,414...
1	2	{"name":"polyline","all_points_x":[415,415,442...
2	3	{"name":"polyline","all_points_x":[415,415,442...
3	4	{"name":"polyline","all_points_x":[443,443,470...
4	5	{"name":"polyline","all_points_x":[388,414,414...
...	...	...
136	137	{"name":"polyline","all_points_x":[63,83,83,63...
137	138	{"name":"polyline","all_points_x":[63,84,84,63...
138	139	{"name":"polyline","all_points_x":[39,60,60,39...
139	140	{"name":"polyline","all_points_x":[39,61,61,39...
140	141	{"name":"polyline","all_points_x":[14,36,36,14...

141 rows × 2 columns

Vehicle_ID entering *Conflict_Zone_ID* at *Time*/ *Frame_ID*

[28]:

	Time	Frame_ID	Vehicle_ID	X_central	Y_central	X1_boundary	Y1_boundary	X3_boundary	Y3_boundary	Conflict_Zone_ID	In_Conflict_Zone
0	0.000000	0	1	323.753850	244.54315	318.220580	239.55382	329.28710	249.53250	-1.0	False
1	0.000000	0	2	391.717830	275.67474	385.785250	270.32385	397.65040	281.02560	-1.0	False
2	0.000000	0	3	401.334100	251.35030	393.590270	244.73557	409.07790	257.96503	1.0	True
3	0.000000	0	4	522.845700	234.48148	511.184140	226.27820	534.50720	242.68474	10.0	True
4	0.033333	1	2	390.745000	275.69020	384.814150	270.33154	396.67587	281.04886	-1.0	False
...	...	...	...	...	...	...	...	...	...	...	...
6615	45.200000	1356	412	55.835953	304.38287	48.771217	298.37247	62.90069	310.39328	140.0	True
6616	45.200000	1356	402	293.596000	285.93573	287.697360	280.71014	299.49463	291.16132	119.0	True
6617	45.200000	1356	357	137.816040	297.97516	131.803470	292.35928	143.82863	303.59103	132.0	True
6618	45.200000	1356	360	759.087300	240.51747	749.301150	234.02133	768.87340	247.01361	-1.0	False
6619	45.200000	1356	413	260.035130	299.60904	255.197220	294.43912	264.87305	304.77896	-1.0	False

6620 rows × 11 columns

entry_time, exit_time and *duration* of a *Vehicle_ID* in *Conflict_Zone_ID*

[18]:

	Vehicle_ID	Conflict_Zone_ID	entry_time	exit_time	duration
35	1	85.0	1.100000	1.533333	0.433333
37	1	2.0	2.533333	2.700000	0.166667
36	1	1.0	1.733333	2.400000	0.666667
34	1	84.0	0.866667	0.900000	0.033333
70	2	121.0	1.933333	1.966667	0.033333
...	...	...	...	...	...
450	410	99.0	44.600000	44.733333	0.133333
451	410	96.0	44.800000	44.900000	0.100000
452	412	140.0	44.666667	45.200000	0.533333
453	413	122.0	44.766667	44.966667	0.200000
454	414	8.0	44.933333	45.033333	0.100000

455 rows × 5 columns

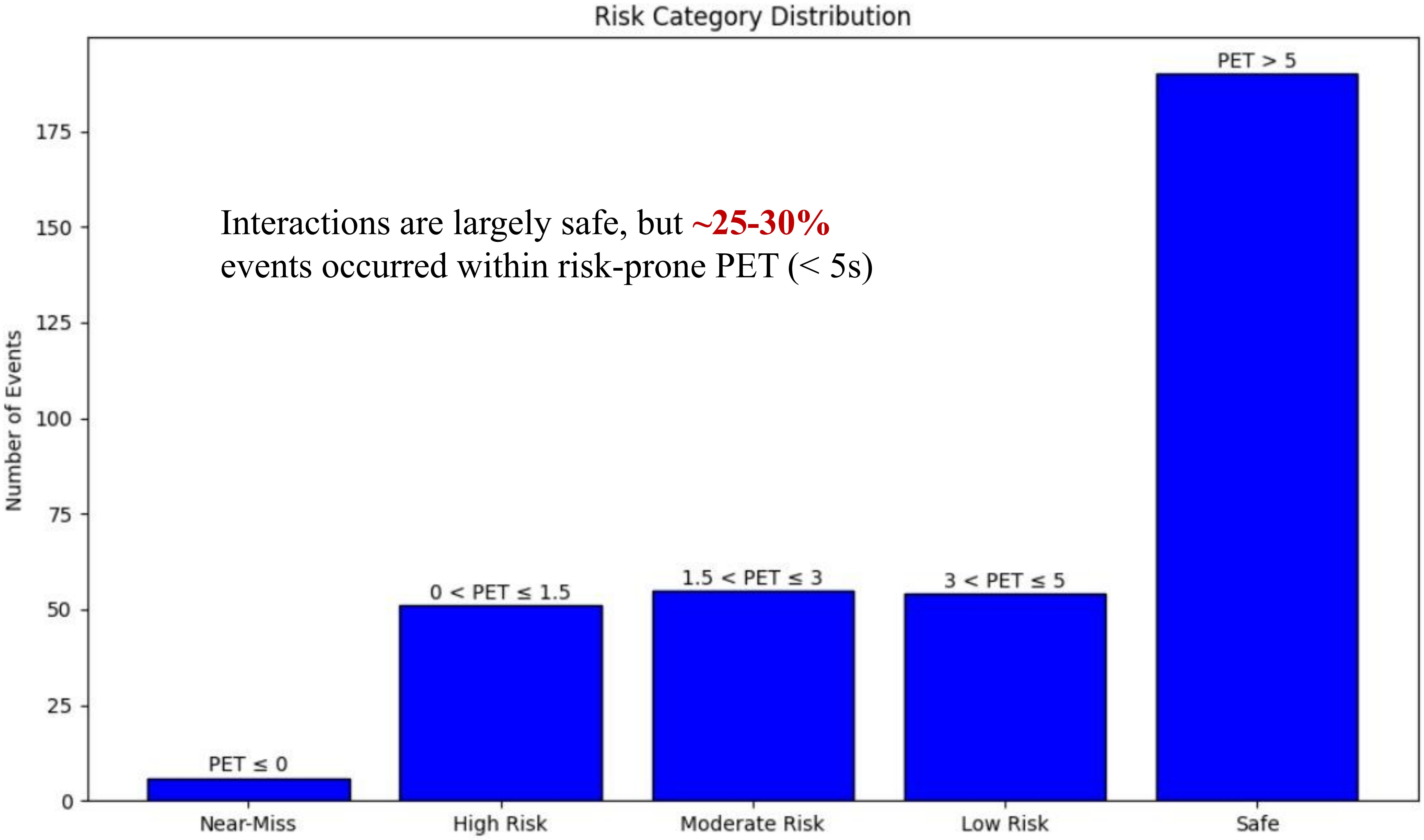
Pet + risk_category value computed for each *zone_id* and vehicle pair entry

[101…

	zone_id	leading_vehicle	following_vehicle	pet	risk_category
0	1.0	31.0	94.0	2.933333	Moderate Risk
1	1.0	94.0	183.0	12.100000	Safe
2	1.0	183.0	234.0	12.733333	Safe
3	2.0	3.0	53.0	2.166667	Moderate Risk
4	2.0	53.0	87.0	4.400000	Low Risk
...	...	...	...	...	...
69	13.0	234.0	382.0	14.600000	Safe
70	13.0	382.0	373.0	-0.100000	Near-Miss
71	14.0	100.0	102.0	0.166667	High Risk
72	14.0	102.0	139.0	6.566667	Safe
73	14.0	139.0	239.0	16.766667	Safe

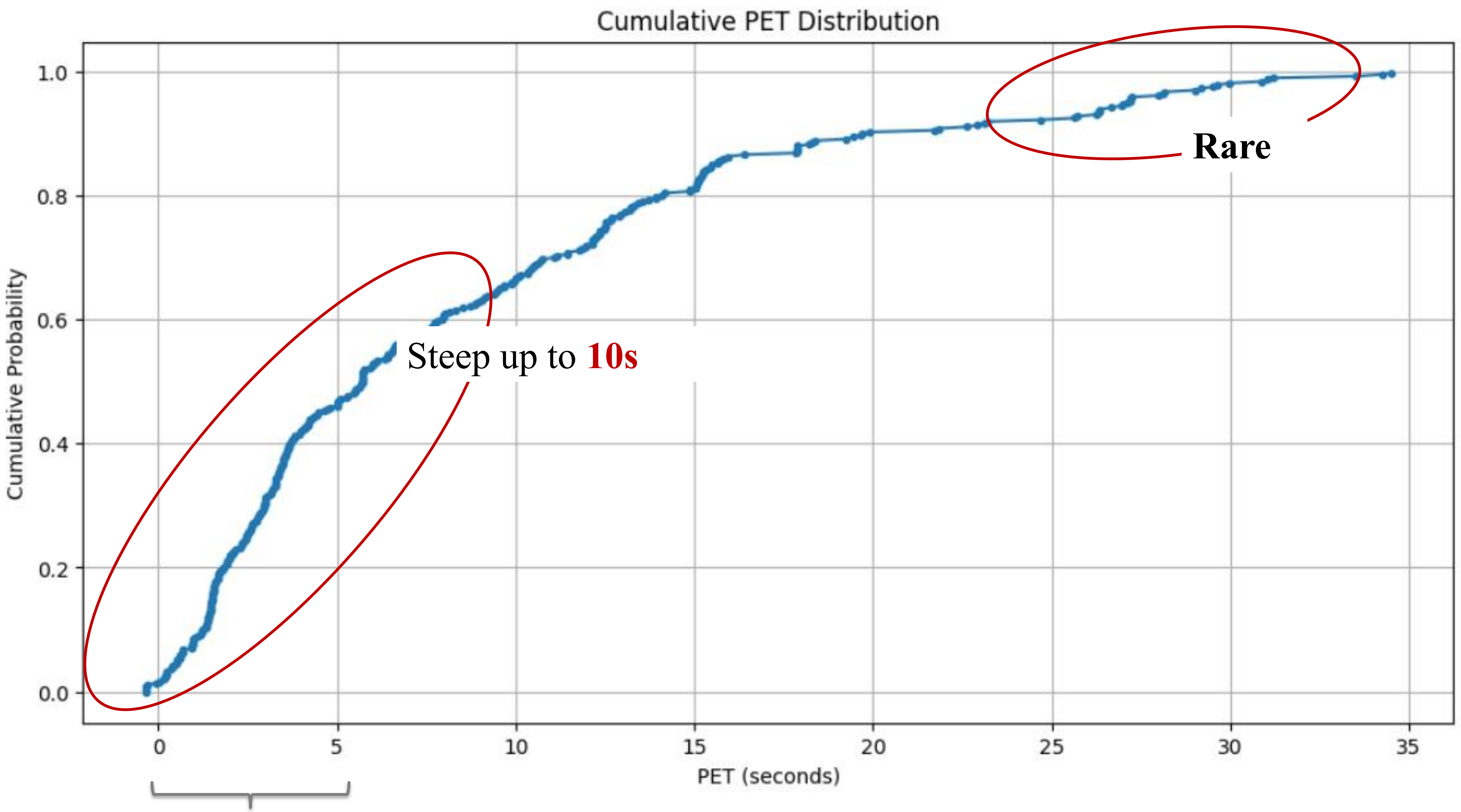
74 rows × 5 columns

Risk Category Distribution bar chart



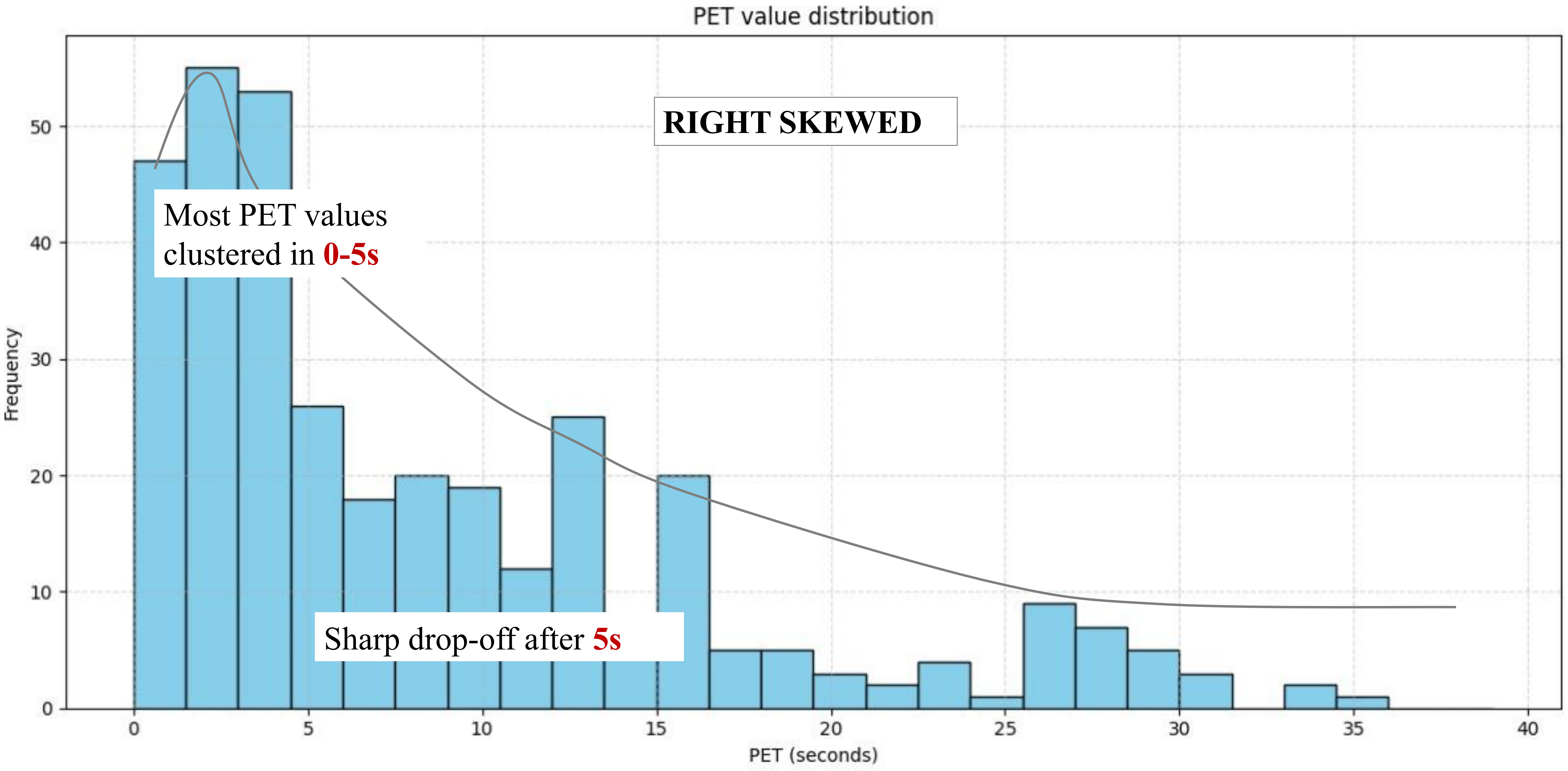
Interactions are largely safe, but **~25-30%** events occurred within risk-prone PET (< 5s)

Cumulative Probability vs PET Distribution



PET $\leq 5s$ occurred in **40-45%** of cases

PET value Hist Plot



Novelty

	Our work	How better than existing work?
1	Our **SDT-ATT** models both who matters and how they move, using **3-neighbor** and **lane-based** behavior, improving prediction in dense traffic and complex scenarios.	Most trajectory models consider **only past trajectory for prediction**, either use pure LSTMs or vanilla social pooling.
2	Added **real-time spatial heatmap** on top.	Prior papers often **do not map PET back to real-world zones dynamically**.
3	**End-to-end integration** (YOLOv8 + ByteTrack + Prediction + Risk Mapping) – fully automated pipeline from raw aerial footage to actionable safety analysis.	Many papers/ repos **isolate these stages**

Ideal in the Indian context

As required in the work scope

Limitations

1 Model performance depended on the number of **annotations** in the drone video.

2 SDTATT assumes **ideal visibility of all neighbors**; real-world occlusions can reduce accuracy.

3 **Lack of clear lane boundaries** in the video reduces lane-change prediction accuracy.

4 Conflict detection is limited to **predicted trajectories without scene semantics** (e.g., road boundaries etc).

5 Lane-change prediction is implicit, **not explicitly modeled or supervised**.

Future Scope of the Project

1. Integrate further **lane and road characteristics** using simple scene semantics to improve trajectory prediction accuracy.

2. Add a dedicated module for **explicit lane-change classification** instead of relying only on inferred motion.

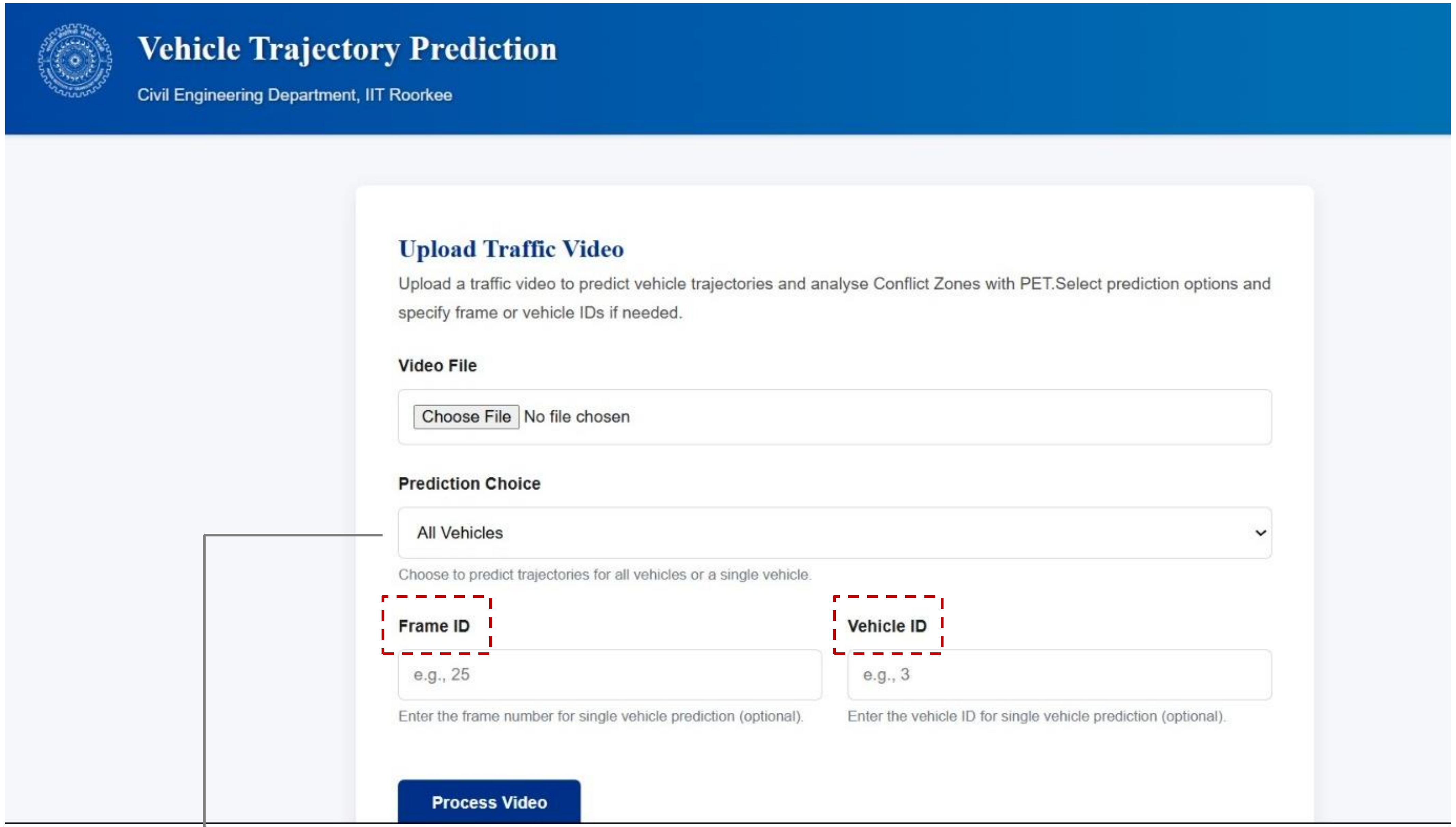
3. Simulate more complex traffic scenarios like **multi-lane roads, intersections, or curved paths** using synthetic datasets

4. Experiment with **Transformer-based models** or lightweight variants to compare performance with LSTMs.

5. Extend object detection to include **non-standard agents** as well, including local agent semantics.

STCAP Platform Demo

SpatioTemporal Trajectory & Collision Analysis Portal (STCAP)



User Input

Aerial drone footage or Tracking.csv

Time, Frame, Vehicle_Id, X,Y (central & boundary coordinates)

Output

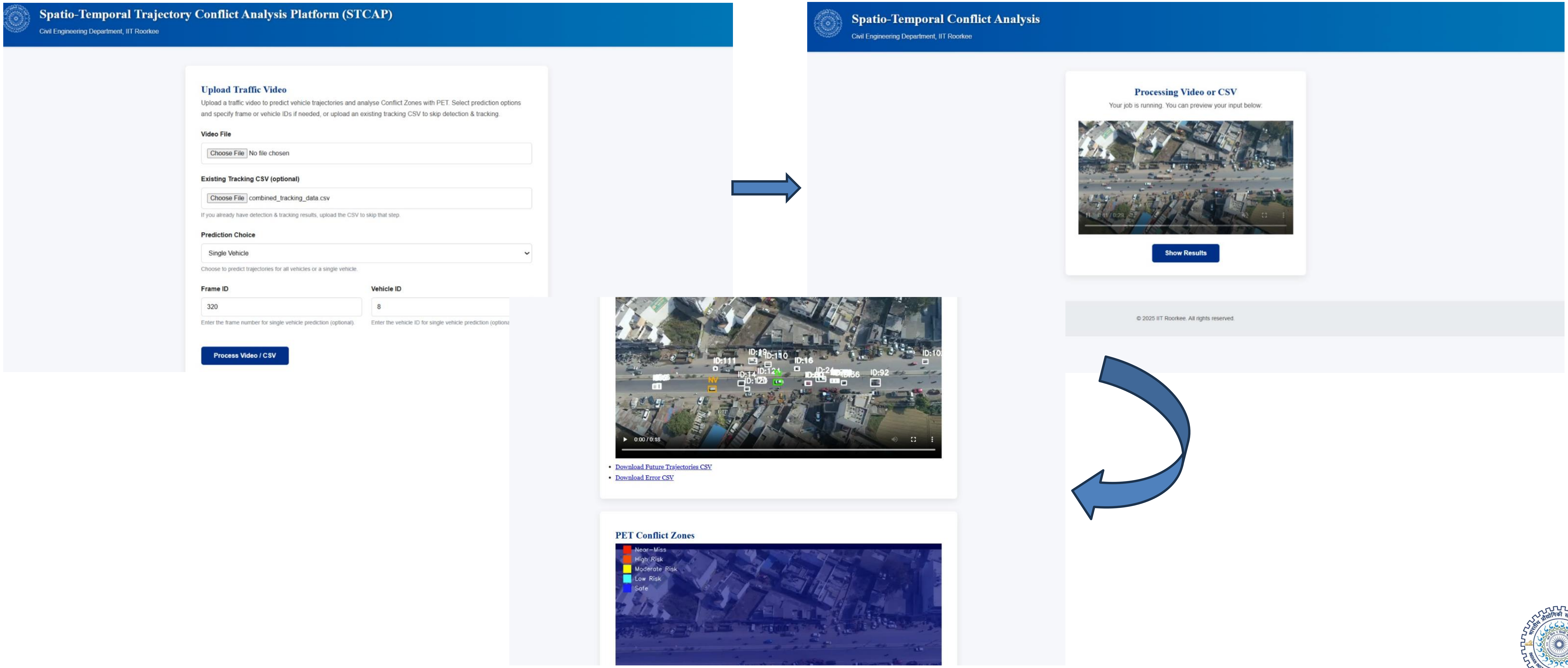
Trajectory Prediction + Lane Changing behavior + Conflict zone mapped (both spatial and temporal)

User can choose to track all vehicles/ or any **specific vehicle** as well

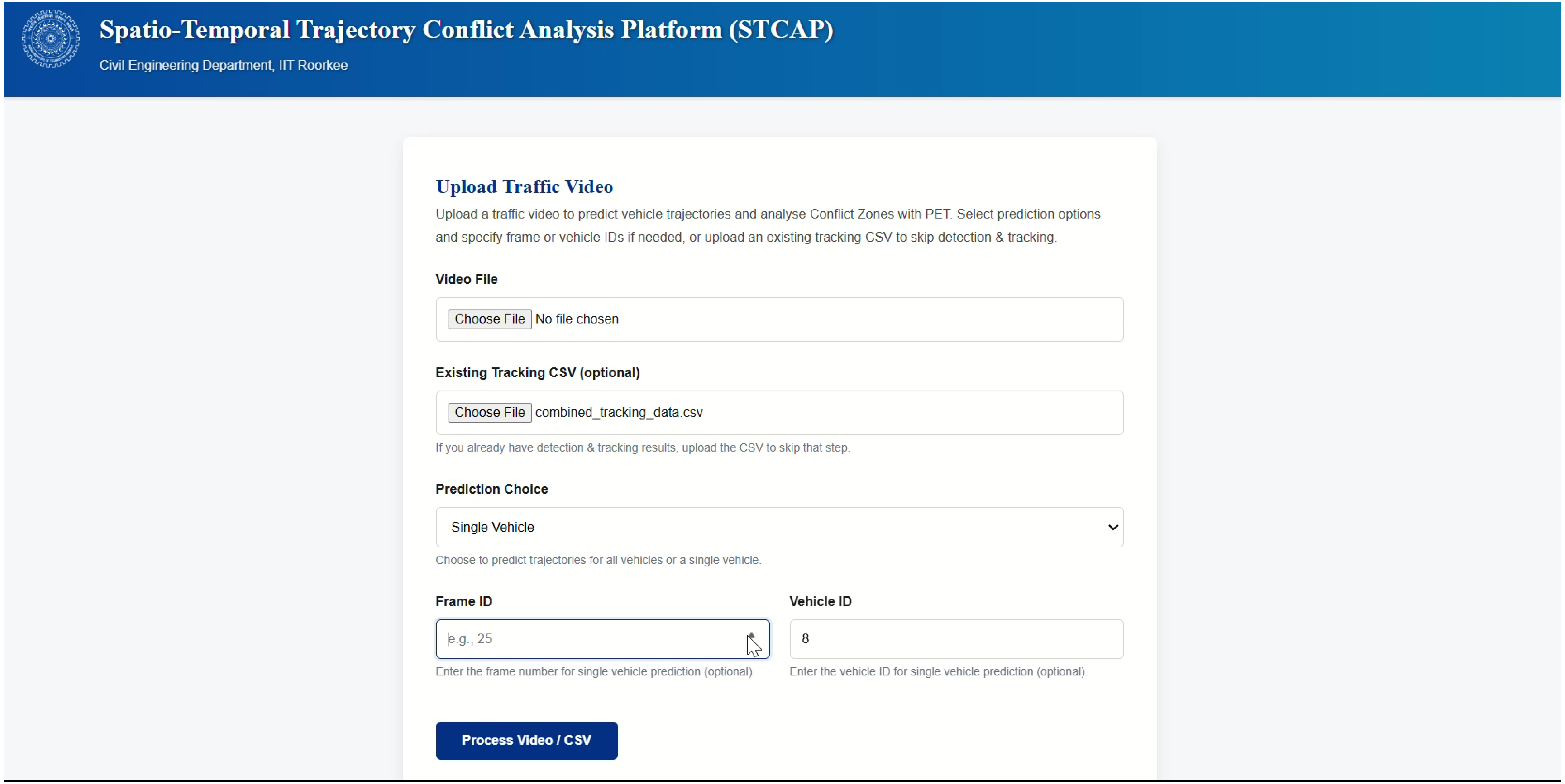
+

User can choose to analyze any **specific conflict zone** as well

Snippets of Platform



Video Demo on Single Vehicle Trajectory and Multiple Vehicle Conflict Zones (PET)



Members Contribution

- ❑ Trajectory Prediction
- ❑ Platform Deployment

- ❑ Conflict zones using PET
- ❑ Report and documentation

- ❑ Data pre-processing
- ❑ Object Detection & Tracking
- ❑ Documentation

- ❑ Literature review
- ❑ Data annotation & augmentation
- ❑ GitHub repo

Thank You